

WESAD-BASED MULTIMODAL EMOTION RECOGNITION USING HYBRID DEEP LEARNING

I. METHODOLOGY AND PREPROCESSING

The WESAD dataset provides a diverse set of physiological measurements collected from both chest-mounted and wrist-worn wearable devices. These recordings include electrocardiogram, respiration, electrodermal activity, temperature, and accelerometer-derived motion signals. The preprocessing pipeline focuses on reconciling differences in sampling rates, synchronizing multimodal streams, and preparing fixed-size windows suitable for deep learning models. Chest signals captured at approximately 700 Hz require downsampling, while wrist recordings at around 32 Hz require upsampling to establish temporal correspondence. This process ensures parallel time alignment without compromising information integrity. Subsequently, acceleration channels from both devices are transformed into magnitude form to reduce dimensional complexity while retaining motion dynamics. After alignment, the channels are concatenated into a unified eight-channel array for each subject. To support temporal modeling, the continuous synchronized data is divided into 10-second windows with a 50% overlap strategy. Majority voting on task labels ensures accurate labeling of windows that lie near transitions between emotional states. Normalization is conducted using statistics computed exclusively from training subjects to maintain proper LOSO evaluation protocol. Given the dataset's inherent imbalance—where baseline samples significantly outnumber stress, amusement, and meditation—the preprocessing pipeline must remain efficient, memory-conscious, and generalizable across subjects.

II. HYBRID MODEL ARCHITECTURE

The proposed architecture integrates convolutional, recurrent, and attention-based components to effectively learn physiological variations corresponding to emotional states. The initial convolutional layers capture local temporal patterns and variations in EDA bursts, respiration cycles, and ECG fluctuations. These low-level features serve as trajectories that feed into a bidirectional GRU, capable of modeling long-term

temporal dependencies essential for physiological interpretation. The bidirectional structure plays a crucial role in capturing both forward and backward dependencies within the time series. Physiological manifestations of stress or amusement often occur as gradual transitions rather than abrupt events. Therefore, the GRU contextualizes each moment using surrounding temporal information, resulting in a more reliable encoding of emotional dynamics. Layer normalization stabilizes the recurrent units and mitigates gradient-related issues during training. To further refine feature extraction, a multi-head self-attention layer is incorporated. This mechanism identifies critical temporal segments that contribute most significantly to emotional discrimination. Instead of uniformly weighting all time steps, the attention layer highlights regions of the signal that contain abrupt stress-induced EDA elevations, amusement-related respiration deviations, or steady meditation patterns. The final classification stage consists of two fully connected layers with dropout for regularization. The model outputs four class predictions representing baseline, stress, amusement, and meditation. Weighted cross-entropy accommodates class imbalance, while AdamW with a learning rate of $1e-3$ ensures stable optimization. Mixed-precision training accelerates throughput, and gradient clipping prevents numerical instability in deeper sequential operations.

III. RESULTS AND DISCUSSION

Experiments are conducted using a Leave-One-Subject-Out strategy to assess generalizability. The model achieves a mean accuracy of 63.1% and a macro F1-score of 30.33%. While baseline detection is relatively strong due to its dominance and distinct physiological characteristics, stress and amusement classification remain challenging. Stress frequently overlaps with baseline in subtle physiological markers, and amusement appears sparsely across subjects, resulting in an F1-score of 0.10. Meditation displays moderate separability with an F1-score of 0.35, likely due to distinct respiration and EDA relaxation patterns. However, the dataset's imbalance continues to skew predictions toward majority classes despite loss weighting. Moreover, inter-subject variability yields a wide performance range across LOSO folds—from 14.82% to 57.63% F1—indicating personalized physiological responses to emotional stimuli. Confusion matrices reveal common misclassifications: stress is frequently interpreted as baseline, and amusement is often misidentified due to subtle signals and limited data

representation. These observations underscore the necessity of additional augmentation techniques, extended window lengths, and advanced temporal modeling strategies. Runtime measurements demonstrate the model's practicality for deployment: training requires roughly 6.5 seconds per epoch on a consumer-grade GPU, and inference completes within 20 milliseconds per window. These characteristics make the architecture well-suited for real-time applications in wearable health monitoring.

IV. CONCLUSION

This research demonstrates an efficient and generalizable approach to emotion recognition using multimodal physiological signals from wearable devices. Through a structured preprocessing pipeline, a hybrid deep neural architecture, and subject-independent evaluation, the system showcases both strengths and limitations inherent in physiological emotion detection. Although challenges persist in distinguishing minority emotional states, the inclusion of attention mechanisms and advanced temporal modeling enhances discriminatory capability. Future work will explore richer augmentation strategies, adaptive loss functions, and more diverse physiological datasets to further improve cross-subject generalizability.

REFERENCES

[1] Schmidt et al., "WESAD: A Multimodal Dataset for Wearable Stress and Affect Detection," 2018.